



Characterization of the existence of an L - U factorization

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Abstract

For the first time, a characterization is given of the circumstances under which an n -by- n matrix over a field has an L - U factorization. This is in terms of a comparison of ranks of the leading k -by- k principal submatrix to the rank of the first k columns and first k rows. Known results about special types of L - U factorizations follow as do some new results about near L - U factorization when a conventional L - U factorization does not exist. The proof allows explicit construction of an L - U factorization when one exists.

Keywords L - U factorization · Leading principal submatrices · Rank

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1 Introduction

An n -by- n matrix $A = (a_{ij})$ over a field $\mathbb{F}$ is said to have an L - U factorization if there exist a lower triangular matrix $L \in \mathcal{M}_n(\mathbb{F})$ and an upper triangular matrix $U \in \mathcal{M}_n(\mathbb{F})$ such that

$$A = LU.$$

No additional requirements are placed upon L or U . L - U factorizations are of both computational/applied importance, as well as theoretical importance in linear algebra

Dedicated to my long time friend, collaborator and co-mentor, Ilya Spitkovsky, whom I value greatly.
(C.J.)

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[2–4]. There are familiar results that characterize when an L - U factorization exists (a) with L invertible, (b) with U invertible [5], or (c) with both L and U invertible [2, 4]. And it may happen that a matrix have an L - U factorization, but none with the above mentioned conditions, or have no L - U factorization of any kind. The matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is the simplest example of the latter sort. Since A is invertible, both L and U would have to be invertible, which quickly leads to a contradiction, as either L or U would have to have its $(1, 1)$ entry 0.

Here, we complete the theory by characterizing those A that have an L - U factorization. Of course, prior mentioned results are special cases, and our result has already been used [1] to show how a matrix may be factored into a product in which the left factor is nearly lower triangular and the right factor is nearly upper triangular. We do not discuss it in detail here, but there is an analogous theory for “ U - L ” factorizations.

2 A general necessary condition for existence

Which $A \in \mathcal{M}_n(\mathbb{F})$, then, do have an L - U factorization? We first observe a new necessary condition on A by writing a supposed factorization in partitioned form. Suppose that

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

in which $A_{11} \in \mathcal{M}_k(\mathbb{F})$, and that

$$A = \begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} U_{11} & U_{12} \\ 0 & U_{22} \end{bmatrix},$$

with the two factors (L and U) partitioned conformally. We then have

$$\begin{bmatrix} A_{11} & A_{12} \end{bmatrix} = L_{11} \begin{bmatrix} U_{11} & U_{12} \end{bmatrix}, \quad \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} = \begin{bmatrix} L_{11} \\ L_{21} \end{bmatrix} U_{11}, \quad \text{and} \quad A_{11} = L_{11} U_{11}.$$

We now use known inequalities involving ranks of products. If $Z = XY$, with X having k columns and Y having k rows, then

$$\text{rank } X + \text{rank } Y - k \leq \text{rank } Z \leq \min\{\text{rank } X, \text{rank } Y\}.$$

Applying first the right hand inequality, we have

$$\text{rank} \begin{bmatrix} A_{11} & A_{12} \end{bmatrix} \leq \text{rank } L_{11} \quad \text{and} \quad \text{rank} \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} \leq \text{rank } U_{11}.$$

Then, applying the left inequality, we get

$$\text{rank } L_{11} + \text{rank } U_{11} - k \leq \text{rank } A_{11}.$$

Combining these, we obtain

$$\text{rank } \begin{bmatrix} A_{11} & A_{12} \end{bmatrix} + \text{rank } \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix} - k \leq \text{rank } A_{11},$$

which we may rewrite as

$$\text{rank } A_{11} + k \geq \text{rank } \begin{bmatrix} A_{11} & A_{12} \end{bmatrix} + \text{rank } \begin{bmatrix} A_{11} \\ A_{21} \end{bmatrix}.$$

This fundamental fact must hold for each $k = 1, \dots, n-1$ (the case $k = n$ is trivial), and, using standard index notation (as found in [4]) here and throughout, these conditions may also be written as

$$\begin{aligned} \text{rank } A[\{1, \dots, k\}] + k &\geq \text{rank } A[\{1, \dots, k\}, \{1, \dots, n\}] \\ &\quad + \text{rank } A[\{1, \dots, n\}, \{1, \dots, k\}], \end{aligned} \quad (2.1)$$

for all $k = 1, 2, \dots, n$.

Note that the mentioned example $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ fails this condition for $k = 1$, as the *LHS* (left hand side) is 1, while the *RHS* (right hand side) is 2, so that this A has no L - U factorization. Note also that what the field is here is unimportant, and that if A has an L - U factorization over an extension field, it must have one over the ground field.

3 Sufficiency of the necessary conditions

We next show that the necessary condition (2.1) is sufficient for L - U factorization over $\mathbb{F}$, giving the desired characterization.

Theorem 3.1 *A matrix $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{F})$ has an L - U factorization if and only if A satisfies conditions (2.1).*

Proof Necessity was proved above. We will prove sufficiency by induction on n . In the case $n = 1$, the theorem is trivially true because any 1-by-1 matrix has an L - U factorization. In particular, if $A = [0]$ we let $L = [0]$ and $U = [0]$. There are many other L - U factorizations of the zero matrix.

Suppose that conditions (2.1) are sufficient for $n \leq m$. We want to prove that conditions (2.1) are also sufficient for $n = m + 1$.

The idea of the proof is as follows. We want to find a lower triangular matrix L and an upper triangular matrix U such that $A = LU$. Suppose

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix},$$

and that

$$A = LU = \begin{bmatrix} l_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} u_{11} & U_{12} \\ 0 & U_{22} \end{bmatrix}$$

partitioned conformally. Block multiplication gives us the equivalent statements:

$$a_{11} = l_{11}u_{11}, \quad (3.1)$$

$$A_{21} = L_{21}u_{11}, \quad (3.2)$$

$$A_{12} = l_{11}U_{12}, \quad (3.3)$$

and

$$A_{22} = L_{21}U_{12} + L_{22}U_{22}. \quad (3.4)$$

We will prove that it is possible to choose the first column of L and the first row of U in such a way that equations (3.1)–(3.3) are satisfied and, additionally, that the $(n-1)$ -by- $(n-1)$ matrix

$$B = A_{22} - L_{21}U_{12}$$

satisfies the conditions (2.1). We call B “a factor complement” of A . (It is always defined. In case $a_{11} \neq 0$, it is the same as the Schur complement of a_{11} in A .) Then, by the inductive hypothesis matrix B has an L - U factorization, so $B = L_0U_0$ with L_0 lower triangular and U_0 upper triangular. We let

$$L_{22} = L_0$$

and

$$U_{22} = U_0.$$

Notice that a matrix L defined this way is lower triangular, and a matrix U defined this way is upper triangular. Also, equations (3.1)–(3.4) are satisfied which implies that $A = LU$.

We will now describe how to choose the first column of L and the first row of U . We need to consider several cases.

Case 1: $a_{11} \neq 0$. In this case we let

$$L[\{1, \dots, n\}, \{1\}] = A[\{1, \dots, n\}, \{1\}]$$

and

$$U[\{1\}, \{1, \dots, n\}] = a_{11}^{-1} A[\{1\}, \{1, \dots, n\}]$$

which gives us

$$\begin{aligned}l_{11} &= a_{11} \\ L_{21} &= A_{21} \\ u_{11} &= 1 \\ U_{12} &= a_{11}^{-1} A_{12}.\end{aligned}$$

Equations (3.1)–(3.3) are obviously satisfied by this choice of the first column of L and first row of U . It remains to prove that the matrix $B = A_{22} - L_{21}U_{12}$ satisfies conditions (2.1).

Let us consider the matrix

$$C = A - \begin{bmatrix} a_{11} \\ L_{21} \end{bmatrix} \begin{bmatrix} 0 & U_{12} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 \\ A_{21} & A_{22} - L_{21}U_{12} \end{bmatrix} = \begin{bmatrix} a_{11} & 0 \\ A_{21} & B \end{bmatrix}. \quad (3.5)$$

To see that C satisfies (2.1), let us fix k ($1 \leq k \leq n$). Then matrix $C[\{1 \dots k\}]$ can be obtained from matrix $A[\{1 \dots k\}]$ by applying several type - 3 elementary column operations. Therefore,

$$\text{rank } C[\{1 \dots k\}] = \text{rank } A[\{1 \dots k\}].$$

In particular,

$$\text{rank } C = \text{rank } A.$$

Similarly,

$$\text{rank } C[\{1 \dots k\}, \{1 \dots n\}] = \text{rank } A[\{1 \dots k\}, \{1 \dots n\}]$$

and

$$\text{rank } C[\{1 \dots n\}, \{1 \dots k\}] = \text{rank } A[\{1 \dots n\}, \{1 \dots k\}].$$

Since A satisfies conditions (2.1) we conclude that C also satisfies conditions (2.1).

From formula (3.5) it now follows that matrix B satisfies conditions (2.1). Indeed, let us fix k ($1 \leq k \leq n - 1$). Because $c_{11} = a_{11} \neq 0$ and $c_{12} = \dots = c_{1,k+1} = 0$ we have

$$\text{rank } C[\{1 \dots n\}, \{2 \dots (k+1)\}] = \text{rank } C[\{1 \dots n\}, \{1 \dots (k+1)\}] - 1.$$

Also because $c_{12} = \dots = c_{1,k+1} = 0$ we see that

$$\text{rank } C[\{1 \dots n\}, \{2 \dots (k+1)\}] = \text{rank } C[\{2 \dots n\}, \{2 \dots (k+1)\}]$$

but $C[\{2 \dots n\}, \{2 \dots k+1\}] = B[\{1 \dots n-1\}, \{1 \dots k\}]$ and therefore

$$\text{rank } B[\{1 \dots (n-1)\}, \{1 \dots k\}] = \text{rank } C[\{1 \dots n\}, \{1 \dots (k+1)\}] - 1. \quad (3.6)$$

Similarly,

$$\text{rank } B[\{1 \dots k\}, \{1 \dots (n-1)\}] = \text{rank } C[\{1 \dots (k+1)\}, \{1 \dots n\}] - 1 \quad (3.7)$$

and

$$\text{rank } B[\{1 \dots k\}] = \text{rank } C[\{1 \dots (k+1)\}] - 1. \quad (3.8)$$

In particular,

$$\text{rank } B = \text{rank } C - 1 = \text{rank } A - 1.$$

Also since C satisfies conditions (2.1) we have

$$\begin{aligned} \text{rank } C[\{1 \dots (k+1)\}] + (k+1) &\geq \text{rank } C[\{1 \dots (k+1)\}, \{1 \dots n\}] \\ &\quad + \text{rank } C[\{1 \dots n\}, \{1 \dots (k+1)\}]. \end{aligned} \quad (3.9)$$

From (3.6)–(3.9) we conclude that

$$\begin{aligned} \text{rank } B[\{1 \dots k\}] + k &\geq \text{rank } B[\{1 \dots k\}, \{1 \dots (n-1)\}] \\ &\quad + \text{rank } B[\{1 \dots (n-1)\}, \{1 \dots k\}]. \end{aligned}$$

So, B satisfies conditions (2.1).

Case 2: $a_{11} = 0$. If $a_{11} = 0$, then conditions (2.1) for $k = 1$ imply that either the first row or the first column of A , or both, are zero. We first consider the case when the first row of A is zero but the first column is not zero. Let i be the smallest integer such that $a_{i1} \neq 0$. We let

$$L[\{1 \dots n\}, \{1\}] = A[\{1 \dots n\}, \{1\}]$$

and

$$U[\{1\}, \{1 \dots n\}] = a_{i1}^{-1} A[\{i\}, \{1, \dots, n\}].$$

Again equations (3.1)–(3.3) are obviously satisfied by this choice of the first column L and the first row of U . It remains to prove that the matrix $B = A_{22} - L_{21}U_{12}$ satisfies conditions (2.1).

Let us consider the matrix

$$C = A - \begin{bmatrix} 0 \\ L_{21} \end{bmatrix} \begin{bmatrix} 0 & U_{12} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ A_{21} & A_{22} - L_{21}U_{12} \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ A_{21} & B \end{bmatrix}. \quad (3.10)$$

Exactly as in Case 1 we prove that matrix C satisfies conditions (2.1), and

$$\text{rank } C = \text{rank } A.$$

From the formula (3.10) it now follows that matrix B satisfies conditions (2.1). Indeed, let us fix k ($1 \leq k \leq n-1$). Because $c_{i1} = a_{i1} \neq 0$ and $c_{i2} = \cdots = c_{i,k+1} = 0$ we have

$$\text{rank } C[\{1 \dots n\}, \{2 \dots (k+1)\}] = \text{rank } C[\{1 \dots n\}, \{1 \dots (k+1)\}] - 1.$$

Also because $c_{12} = \cdots = c_{1,k+1} = 0$ we see that

$$\text{rank } C[\{1 \dots n\}, \{2 \dots (k+1)\}] = \text{rank } C[\{2 \dots n\}, \{2 \dots (k+1)\}],$$

but

$$\text{rank } C[\{2 \dots n\}, \{2 \dots (k+1)\}] = \text{rank } B[\{1 \dots (n-1)\}, \{1 \dots k\}],$$

and therefore

$$\text{rank } B[\{1 \dots (n-1)\}, \{1 \dots k\}] = \text{rank } C[\{1 \dots n\}, \{1 \dots (k+1)\}] - 1.$$

If $k+1 \geq i$, similar reasoning gives

$$\text{rank } B[\{1 \dots k\}, \{1 \dots (n-1)\}] = \text{rank } C[\{1 \dots (k+1)\}, \{1 \dots n\}] - 1$$

and

$$\text{rank } B[\{1 \dots k\}] = \text{rank } C[\{1 \dots (k+1)\}] - 1.$$

In particular,

$$\text{rank } B = \text{rank } C - 1 = \text{rank } A - 1.$$

But if $k+1 < i$, we have

$$\text{rank } B[\{1 \dots k\}, \{1 \dots (n-1)\}] = \text{rank } C[\{1 \dots (k+1)\}, \{1 \dots n\}]$$

and

$$\text{rank } B[\{1 \dots k\}] = \text{rank } C[\{1 \dots (k+1)\}].$$

Also since C satisfies conditions (2.1) we have

$$\begin{aligned} \text{rank } C[\{1 \dots (k+1)\}] + (k+1) &\geq \text{rank } C[\{1 \dots (k+1)\}, \{1 \dots n\}] \\ &\quad + \text{rank } B[\{1 \dots (n-1)\}, \{1 \dots k\}]. \end{aligned}$$

In other words B satisfies conditions (2.1).

Notice that A^T has an L - U factorization if and only if A does, as $A = LU$ implies $A^T = U^T L^T = L' U'$, as $U^T = L'$ is lower ($L^T = U'$ is upper) triangular. This means

that the case in which the first column of A is 0 and the first row is not 0 is implied by the case just studied, as the rank conditions are analogous.

The last case we have to consider is when both the first row and the first column of A are equal to zero. If $A = 0$, it may be written $A = 0 \cdot 0$, which is an L - U factorization. If A is not the zero matrix let i be the smallest integer such that the i th column of A is not zero. Let us consider the matrix C such that the first column of C is equal to the i th column of A and columns 2 through n of C are equal to the corresponding columns of A . We want to prove that C satisfies conditions (2.1). Fix k ($1 \leq k \leq n$). If k is less than i , then because the first row of C is zero we have

$$\text{rank } C[\{1 \dots k\}, \{1 \dots n\}] \leq k - 1$$

and because columns 2 through k of C are zero, but the first column of C is not zero we have

$$\text{rank } C[\{1 \dots n\}, \{1 \dots k\}] = 1.$$

We conclude that

$$\text{rank } C[\{1 \dots k\}] + k \geq \text{rank } C[\{1 \dots k\}, \{1 \dots n\}] + \text{rank } C[\{1 \dots n\}, \{1 \dots k\}],$$

which means that C satisfies condition (2.1) for index k .

If $k \geq i$, then because columns 2 through n of C are equal to the corresponding columns of A , and the first column of C is equal to the i th column of A and also because the first column of A is zero, we have

$$\text{rank } C[\{1 \dots k\}] = \text{rank } A[\{1 \dots k\}].$$

In particular,

$$\text{rank } C = \text{rank } A.$$

Also,

$$\text{rank } C[\{1 \dots k\}, \{1 \dots n\}] = \text{rank } A[\{1 \dots k\}, \{1 \dots n\}]$$

and

$$\text{rank } C[\{1 \dots n\}, \{1 \dots k\}] = \text{rank } A[\{1 \dots n\}, \{1 \dots k\}].$$

Since A satisfies conditions (2.1) for index k , we conclude that C satisfies conditions (2.1) for index k . We already know that because C satisfies conditions (2.1) and the first row of C is zero, but the first column is not zero, there are scalars m_{11} , v_{11} , a column vector M_{21} , and a row vector V_{12} such that

$$c_{11} = 0 = m_{11} v_{11},$$

$$\begin{aligned}C_{21} &= M_{21}v_{11}, \\C_{12} &= A_{12} = m_{11}V_{12},\end{aligned}$$

and additionally that the matrix

$$C_{22} - M_{21}V_{12}$$

satisfies conditions (2.1) and

$$\text{rank}(C_{22} - M_{21}V_{12}) = \text{rank } C - 1 = \text{rank } A - 1.$$

We now let

$$\begin{aligned}l_{11} &= m_{11} \\L_{21} &= M_{21} \\u_{11} &= 0 \\U_{12} &= V_{12}.\end{aligned}$$

Equations (3.1)–(3.3) are obviously satisfied by this choice of first column of L and the first row of U . Additionally, a factor complement of A

$$B = A_{22} - L_{21}U_{12} = C_{22} - M_{21}V_{12}$$

satisfies conditions (2.1) and also

$$\text{rank } B = \text{rank}(C_{22} - M_{21}V_{12}) = \text{rank } A - 1.$$

Note that this case can also be treated by considering matrix C such that the first row of C is equal to the first nonzero row of A and such that rows 2 through n of C are equal to the corresponding rows of A . $\square$

We note that, when an L - U factorization exists, following the proof of sufficiency just given, provides an explicit algorithm to produce an L - U factorization (which may be highly non-unique).

4 Some applications

Our Theorem 3.1 may be used to reproduce the classical results about L - U factorization (with L or U or L and U invertible) mentioned in the introduction. For example,

Corollary 4.1 *If $A \in \mathcal{M}_n(\mathbb{F})$ is invertible, then A has an L - U factorization (of course, with L and U invertible) if and only if $\text{rank } A[\{1 \dots k\}, \{1 \dots n\}] = \text{rank } A[\{1 \dots n\}, \{1 \dots k\}] = k$ for $k = 1, \dots, n$.*

Proof Since A is invertible, we have

$$\text{rank } A[\{1 \dots k\}, \{1 \dots n\}] = \text{rank } A[\{1 \dots n\}, \{1 \dots k\}] = k$$

for $k = 1, \dots, n$. From Theorem 3.1, then, the stated condition is necessary and sufficient in this case. $\square$

What, then, if conditions (2.1) fails? Of course, there is no L - U factorization, but how “close” may we come?

Definition 4.2 Matrix $K = (k_{ij}) \in \mathcal{M}_n(\mathbb{F})$ is said to be *almost lower triangular*, with m extra diagonals, if $k_{ij} = 0$ whenever $j > i + m$. An *almost upper triangular matrix*, with m extra diagonals, is defined similarly (or via transposition).

Definition 4.3 We say that $A \in \mathcal{M}_n(\mathbb{F})$ fails conditions (2.1) by at most m if the $LHS + m \geq RHS$ for all k .

Now, for $A \in \mathcal{M}_n(\mathbb{F})$, let $\tilde{A}_m \in \mathcal{M}_{n+m}(\mathbb{F})$ be the result of appending m 0 columns to A on the left and m 0 rows on the top. A simple observation is that

Lemma 4.4 If $A \in \mathcal{M}_n(\mathbb{F})$, then $\tilde{A}_m$ satisfies conditions (2.1) if and only if A fails conditions (2.1) by no more than m .

With this, we may then give a factorization result that applies to any matrix.

Theorem 4.5 Matrix $A \in \mathcal{M}_n(\mathbb{F})$ may be written as $A = KW$, in which K is an almost lower triangular matrix with m extra diagonals and W is an almost upper triangular matrix with m extra diagonals if and only if A fails conditions (2.1) by at most m .

Proof Given an n -by- n matrix A that fails conditions (2.1) by no more than m we obtain the $(m+m)$ -by- $(n+m)$ matrix $\tilde{A}_m$ that satisfies conditions (2.1). We apply Theorem 3.1 to obtain an L - U factorization of $\tilde{A}_m$. As was noted above, the first m rows and the last m columns of L are 0, as well as the first m columns and the last m rows of U are 0. We let

$$K = L[\{(1+m) \dots (n+m)\}, \{1 \dots n\}]$$

and

$$W = U[\{1 \dots n\}, \{(1+m) \dots (n+m)\}].$$

Matrices K and W defined this way have the desired form. Also block multiplication shows that

$$A = KW.$$

For necessity, suppose A can be written as

$$A = KW$$

with K and W as indicated. We define $\tilde{A}_m$, $\tilde{L}_m$, and $\tilde{U}_m$ and block multiplication shows that

$$\tilde{A}_m = \tilde{K}_m \tilde{W}_m.$$

Therefore, by Theorem 3.1, $\tilde{A}_m$ satisfies conditions (2.1), and A fails conditions (2.1) by no more than m , because of Lemma 4.4. $\square$

Example 4.6 The matrix

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

has no L - U factorization, as was noted. However, it may be written as

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

in which the second factor is upper triangular with 1 extra diagonal, or as

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

in which the first factor is lower triangular with 1 extra diagonal, both consistent with Theorem 4.5.

As the example suggests, we may often do better than Theorem 4.5 guarantees in such quasi- L - U factorizations. The extra diagonals need not be full or may not be present in both factors, etc. This was investigated in some cases, using our ideas in [1].

Another variation that applies to general matrices is the following. Now, we only add rows or columns instead of rows and columns to the factors and do not change the matrix to be factored.

Definition 4.7 An n -by- $(n+m)$ matrix H is said to be *almost lower triangular with m extra columns* if the matrix $H[\{1, \dots, n\}, \{(1+m), \dots, (n+m)\}]$ is lower triangular. An $(n+m)$ -by- n matrix W is said to be *almost upper triangular with m extra rows* if the matrix $W[\{(1+m), \dots, (n+m)\}, \{1, \dots, n\}]$ is upper triangular.

We now have (with an analogous proof)

Theorem 4.8 Matrix $A \in \mathcal{M}_n(\mathbb{F})$ may be written as

$$A = HV$$

in which H is almost lower triangular with m extra columns and V is almost upper triangular with m extra rows if and only if A fails conditions (2.1) by no more than m .

The classical fact that

Theorem 4.9 Any matrix $A \in \mathcal{M}_n(\mathbb{F})$ can be rewritten

$$A = PLU,$$

in which $U \in \mathcal{M}_n(\mathbb{F})$ is upper triangular, $L \in \mathcal{M}_n(\mathbb{F})$ is lower triangular, and P is a permutation matrix.

may also be deduced from Theorem 3.1. The permutation (with different factors, L , U , P) may also be placed on the right.

Two further factorizations of any $A \in \mathcal{M}_n(\mathbb{F})$ may also be deduced from Theorem 3.1.

Theorem 4.10 Any matrix $A \in \mathcal{M}_n(\mathbb{F})$ may be written as

$$A = U_1 L U_2$$

or

$$A = L_1 U L_2$$

in which L , L_1 , and $L_2 \in \mathcal{M}_n(\mathbb{F})$ are lower triangular and U_1 , U_2 , and $U \in \mathcal{M}_n(\mathbb{F})$ are upper triangular.

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